

Beyond Geometric Preorganization: A Bottleneck-Guided Pre-Reaction-State Framework for PET Hydrolase Engineering

Jiayi Li, Nianyi Cheng, Xiaofan Luo, Jiayu Wang, Atita Sartsasi, Qingxin Ma, Chenming Jin, Siqi Ding, Ratchakrit Chumchuwat, Flora Da, Yifan Chen, Xinglin Zhang, Haoran Luo, Maoyuan Sun, Ziyu Zhong, Bohang Gan, Yueyang Du, Jiahua Wang, Yishan Gao, Peilin Zhu

Abstract

Enzyme engineering can identify PET hydrolase variants with improved activity, yet the catalytic bottleneck relieved by a beneficial mutation often remains unclear. Here we compare two experimentally validated FAST-PETase substitutions discovered through distinct design routes. Graph-learning-guided Y87Q preserves the global fold while remodeling the second-shell environment and increasing the population of geometric pre-reaction states (PRSs). Molecular dynamics and distortion/interaction analysis associate this redistribution with a reduced structural cost of reaching the transition-state region. By contrast, the active-site substitution S187Y enhances PET-film hydrolysis in two enzyme scaffolds despite weak correspondence between docking score and activity, and quantum-chemical analysis indicates enhanced transition-state interactions. These results motivate a two-level framework separating PRS quantity, the accessibility of attack-compatible configurations, from PRS quality, their conditional chemical productivity. This bottleneck-guided framework links conformational sampling with chemical reactivity and provides a mechanistic basis for iterative PET hydrolase engineering.

Keywords: Poly (ethylene terephthalate), PET hydrolase, Molecular dynamics simulation, Density functional theory, Pre-reaction state

1. Introduction

Poly (ethylene terephthalate) (PET) is a major synthetic polyester whose chemical durability also contributes to persistent plastic waste. Enzymatic depolymerization provides a complementary route to mechanical and thermochemical recycling because PET hydrolases can release soluble products under comparatively mild conditions [1-3]. Structure-guided design, directed evolution and machine-learning-assisted engineering have produced increasingly active and thermostable PET hydrolases, including FAST-PETase and HotPETase [4,5]. Yet a central challenge remains: beneficial mutations are often identified faster than the physical bottlenecks they relieve, limiting the transfer of successful designs across enzyme scaffolds [6,7].

This mechanistic problem is amplified by the heterogeneous, interfacial nature of PET hydrolysis. Productive turnover can depend on polymer-surface association, entry of a PET chain into the binding cleft, formation of a catalytically competent bound state and the subsequent chemical steps [8,9]. Consequently, macroscopic PET-film degradation cannot be assigned to a single microscopic descriptor. Binding scores, global stability and even one optimized enzyme-substrate structure may therefore fail to reveal whether a mutation acts by improving substrate access, conformational sampling or the chemistry of bond cleavage. A useful design framework should separate these contributions rather than treating activity as the direct consequence of one static structural feature.

PET hydrolysis by serine hydrolases requires more than substrate binding. During acylation, the catalytic serine must approach the scissile ester carbonyl with an appropriate nucleophile-electrophile distance and Bürgi-Dunitz angle, while the carbonyl oxygen is oriented toward the oxyanion hole and the catalytic His-Asp network supports proton transfer [10,11]. Near-attack-conformation and pre-reaction-state (PRS) analyses therefore provide a physically interpretable description of reaction-ready configurations sampled within enzyme-substrate ensembles [12,13]. Related studies have further emphasized the importance of substrate positioning, conformational dynamics and active-site organization in enzyme catalysis [14].

A purely geometric definition of reaction readiness is nevertheless incomplete. Two configurations may satisfy the same attack-distance and angle criteria yet differ in oxyanion-hole engagement, proton-transfer readiness, electrostatic organization, substrate distortion or transition-state stabilization. Conversely, a mutation may increase catalytic output mainly by raising the probability of entering an attack-compatible geometry, even if the intrinsic reactivity of each such configuration changes only modestly. This distinction is especially relevant to computational enzyme engineering, because machine-learning models, docking calculations and physics-based methods optimize different surrogate objectives [15].

Here we compare two experimentally validated FAST-PETase substitutions identified through distinct design routes. Y87Q, obtained through graph-learning-guided design, enhances PET depolymerization while redistributing the bound ensemble toward geometric PRSs. S187Y, identified through active-site design and screening, improves PET-film hydrolysis in two enzyme backgrounds despite poor correspondence between docking scores and experimental activity. By integrating experiment, molecular dynamics, PRS analysis and quantum-chemical distortion/interaction analysis, we show that the two substitutions act through distinguishable physical routes. We therefore propose a two-level framework in which PRS quantity describes access to attack-compatible configurations, whereas PRS quality describes their conditional chemical productivity. This distinction provides a bottleneck-oriented basis for matching enzyme-engineering strategies to the limitation that remains in a given scaffold.

2. Results

2.1 A two-stage design strategy identifies Y87Q as an activity-enhancing FAST-PETase variant

We established a two-stage computational–experimental workflow that couples structure-aware graph learning with experiment-informed refinement (Fig.1A). In the first stage, an optimized geometric vector perceptron graph neural network (GVP-GNN) explored local mutational space while retaining three-dimensional structural information. Candidate substitutions were ranked by residue-level amino-acid probabilities and filtered according to their proximity to the catalytic pocket or docked PET-derived substrate. Experimental measurements were then returned to a second stage combining fine-tuned ESM2 embeddings, learn2learn-based meta-learning and Gaussian-process regression to refine the sequence–function landscape. This design architecture follows our graph-learning/meta-learning strategy for PET hydrolase engineering while explicitly connecting candidate generation to subsequent mechanistic analysis.

Y87Q emerged as the strongest FAST-PETase candidate from the first-stage screen. To emphasize performance at elevated temperature, candidate activity was summarized using a temperature-weighted score. Y87Q reached a normalized score of 4.32, compared with 2.17 for the parental enzyme (Fig.1B). We then tested purified enzymes against amorphous glycol-modified PET films over 40–70 °C. Y87Q released approximately 1.9-fold more soluble hydrolysis products than FAST-PETase at 60 °C and 4.9-fold more at 70 °C (Fig. 1c,d). Time-course measurements at 60 °C likewise showed faster product accumulation (Fig. 1E,F).

Together, these experiments established Y87Q as an activity-enhancing substitution and provided a defined phenotype for mechanistic analysis. The improvement was expressed primarily as greater and faster soluble-product formation rather than an evident qualitative change in product identity. Because Y87Q lies outside the catalytic triad, this phenotype raised a specific mechanistic question: whether the mutation improves catalysis through global stabilization, stronger substrate binding or redistribution of the bound conformational ensemble.

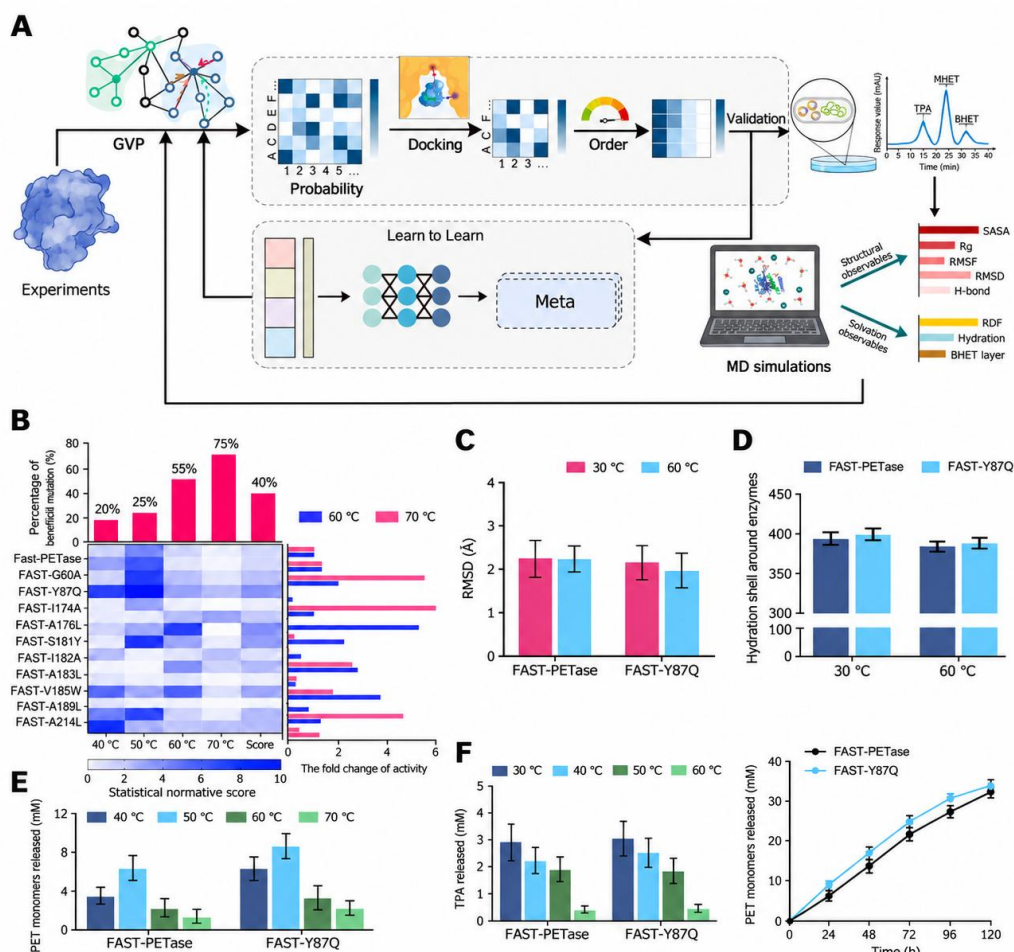


Fig.1 Graph-learning-guided identification and experimental validation of Y87Q.

(A) Two-stage computational–experimental workflow for PET hydrolase engineering. (B) Temperature-dependent screening and weighted activity scores. (C) (D) Backbone RMSD and hydration analysis from MD simulations. (E) PET hydrolysis activity at different temperatures. (F) TPA release and time-dependent accumulation of soluble PET hydrolysis products.

2.2 Y87Q relieves a geometric-accessibility bottleneck through second-shell remodeling

Molecular dynamics simulations were next used to determine whether Y87Q acts through global stabilization or through local reorganization of the bound catalytic ensemble. At 30 and 60 °C, the mutant retained a backbone RMSD comparable to that of FAST-PETase, and the overall hydration-shell population changed little. The dominant structural consequences were instead localized near the active-site environment, arguing against a large-scale reconstruction of the enzyme fold.

A more persistent structural signature emerged from the local second-shell interaction network (Fig.2A). In the Y87Q simulations, Gln87 formed a directional contact with Met161, with the Gln87 carbonyl oxygen approaching the Met161 sulfur at an S···O separation of approximately 3.0 Å and with a C–S···O geometry consistent with interaction along the extension of the C–S bond [16]. Electronic-structure analyses supported the presence of a weak but directional closed-shell interaction. Natural bond orbital analysis identified donation from the Gln87 carbonyl-oxygen lone pair into the $\sigma^*(\text{C–S})$ orbital of Met161, while quantum theory of atoms in molecules analysis identified an S···O bond path. Together with the corresponding IGMH analysis, these results support a locally reorganized second-shell interaction network associated with the Y87Q substitution.

We then asked whether this local remodeling changes the probability of reaching an attack-compatible configuration. A geometric PRS was defined using the Ser160-O_γ···C_{acyl} distance and Bürgi–Dunitz approach angle, consistent with the stereochemical requirements of nucleophilic addition to a carbonyl center [17]. FAST-PETase occupied this PRS in $11.9 \pm 1.4\%$ of sampled configurations, whereas Y87Q reached $24.2 \pm 2.8\%$ (Fig.2B,C). This approximately twofold enrichment was accompanied by shorter nucleophile–electrophile separations and more favorable attack angles. The Gln87–Met161 contact showed an S···O separation of approximately 3.0 Å and a geometry compatible with chalcogen-bond-like interactions [18]. NBO analysis further identified lone-pair donation from the Gln carbonyl oxygen into the $\sigma^*(\text{C–S})$ orbital of Met161, with a second-order stabilization energy of 2.9 kcal mol⁻¹, while QTAIM identified the corresponding S···O bond path (Fig.2D–F). Additional solvation, flexibility and IGMH analyses are reported in the Supplementary Information.

Reaction-ready configurations also more frequently coincided with engagement of the oxyanion-hole region, indicating that the observed redistribution was not limited to one distance coordinate. Extended trajectories supported persistent changes in local solvation and maintenance of a shorter substrate–catalytic-serine separation. These results identify the principal effect of Y87Q as an increase in geometric accessibility.

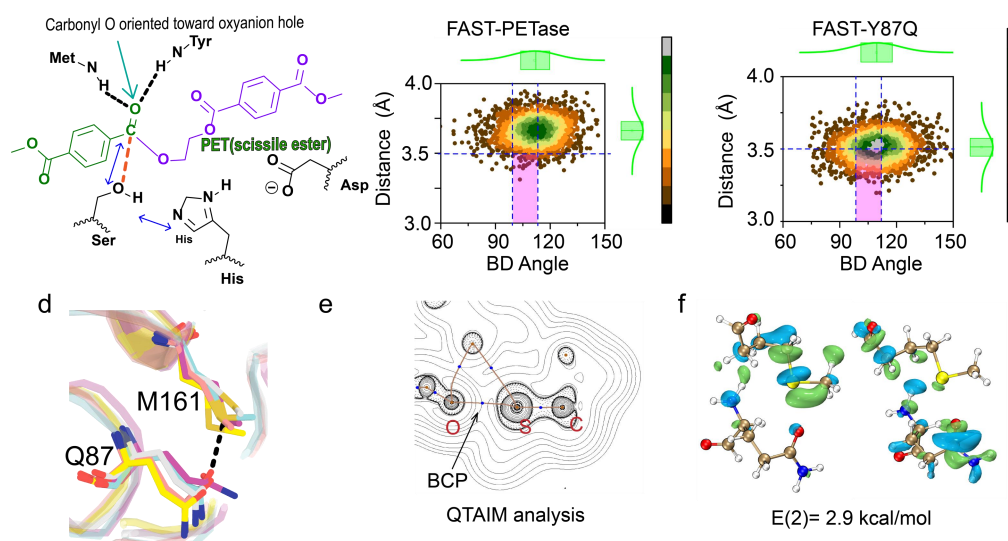


Fig.2 Y87Q remodels the second-shell environment and increases PRS sampling.

(A) Reaction geometry used to define the pre-reaction state (PRS). (B)(C) Joint distributions of the Ser160-O_y...C_{acyl} distance and Bürgi–Dunitz angle for FAST-PETase and FAST-Y87Q. (D) Representative Gln...Met161 interaction. e,f, QTAIM and NBO analyses of this interaction.

2.3 S187Y enhances PET hydrolysis through an active-site mechanism not captured by docking affinity

To test whether improved PET hydrolysis can arise through a distinct route, we examined S187Y, an active-site substitution identified through an independent structure-based workflow (Fig.3A). Residues surrounding the substrate-binding region were enumerated computationally, candidate structures were refined and the resulting variants were filtered by docking to a PET-derived ligand. Nine top-ranked single substitutions were then evaluated experimentally. This strategy provided a complementary test because docking primarily scores bound-state complementarity rather than the complete catalytic trajectory.

PET-film hydrolysis was measured at 40 °C over 72 h using absorbance at 260 nm together with UPLC quantification of MHET and TPA. S187Y produced the largest activity increase among the tested variants, yielding a 2.18-fold enhancement relative to the parental FAST-PETase-derived scaffold (Fig.3B–D). Importantly, experimental activity did not track the docking-energy ranking: several variants with favorable docking scores showed little or no improvement. Thus, stabilization of a docked ground-state pose was insufficient to explain the catalytic phenotype. The strong residue dependence at this position further supports this conclusion: alternative substitutions at the same site produced markedly different activities even when their docking scores were favorable.

The S187Y effect was also transferable. Introducing the same substitution into the independently engineered ThermoStable-PETase scaffold increased PET-film degradation by 3.44-fold, and UPLC confirmed increased formation of both MHET and TPA. This cross-scaffold activity enhancement suggests that S187Y modifies a catalytic feature shared by the related active-site architectures rather than producing only a narrowly scaffold-specific packing effect. The complete variant comparison and concentration-dependent behavior are presented in the Supplementary Information.

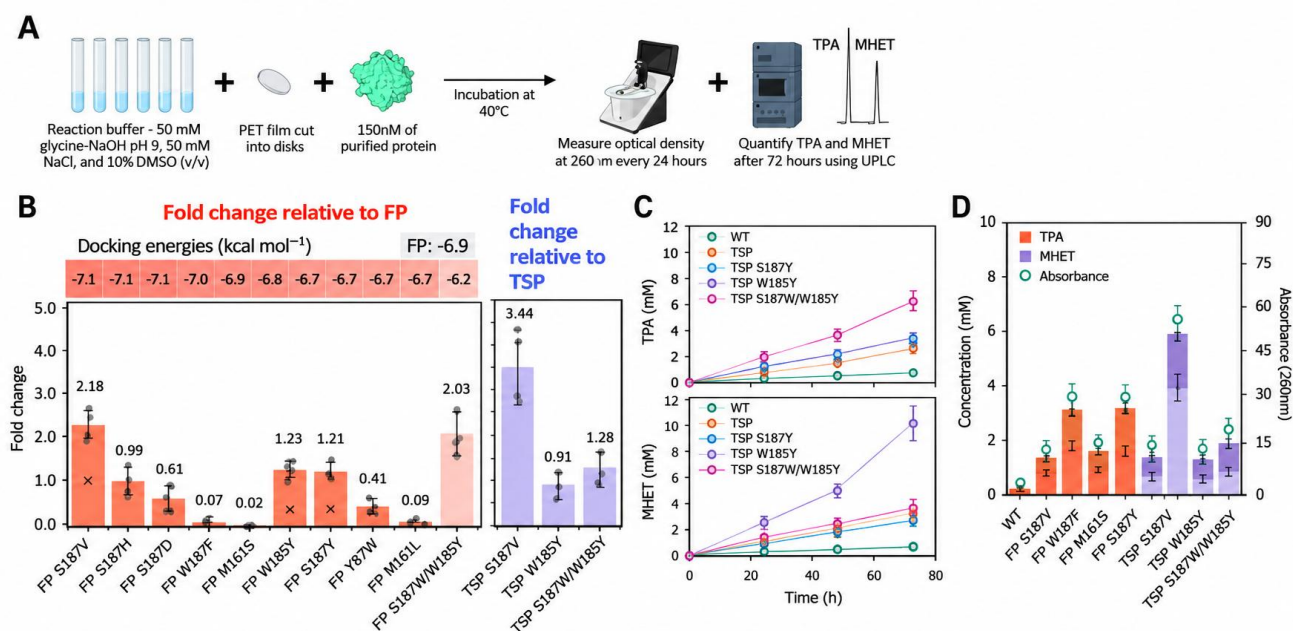


Fig.3 Identification and experimental validation of the activity-enhancing S187Y substitution.

(A) PET-film hydrolysis workflow. (B) Relative activities and docking energies of active-site variants in the FP and TSP scaffolds. (C) Time-dependent accumulation of PET hydrolysis products. (D) UPLC quantification of TPA and MHET after 72 h.

2.4 Quantum-chemical analysis separates conformational preorganization from transition-state stabilization

Representative structures from the geometric PRS ensembles were subjected to a common all-QM cluster protocol to examine the acylation step. In all three systems, the reaction retained the canonical serine-hydrolase topology: Ser160 attacks the PET ester carbonyl, proton transfer is coupled to His237, and

developing oxyanion character is stabilized by the local hydrogen-bond network [19-20]. We found no indication that either beneficial substitution creates an alternative reaction pathway. Instead, the variants altered the energetic cost of reaching and stabilizing the same transition-state region.

The mean calculated acylation barriers were 20.1, 18.6, 17.8 kcal mol⁻¹ for FAST-PETase, FAST-Y87Q, and FAST-S187Y, respectively (Fig.4A). Despite these comparable barrier reductions, distortion/interaction activation-strain (DIAS) analysis revealed different energetic origins (Fig.4B). In FAST-Y87Q, the improvement was dominated by a reduced distortion penalty relative to FAST-PETase, consistent with its enrichment of attack-compatible PRS geometries. Instead, the mutations change how efficiently the enzyme reaches and stabilizes the same transition-state region. Y87Q primarily reduces the conformational and distortion cost of arriving there, whereas S187Y gains more from favorable interactions once a reaction-ready configuration has formed (Fig.4C).

The use of multiple PRS-derived QM starting structures is important because the mechanistic comparison concerns distributions, not one optimized snapshot. The barrier distributions shift for both beneficial variants, but DIAS resolves why those shifts occur. A lower distortion requirement is consistent with improved preorganization, whereas a more favorable interaction term is consistent with stronger stabilization in the transition-state region. Reading the two terms together avoids assigning every lower barrier to the same structural cause and provides an energetic bridge between ensemble sampling and chemical reactivity.

Because the mechanistic comparison concerns conformational ensembles rather than a single optimized snapshot, multiple PRS-derived starting structures were used for the quantum-chemical calculations. The resulting barrier distributions shifted for both beneficial variants, while DIAS analysis resolved distinct energetic origins for these shifts. This ensemble-based treatment is important for PET hydrolases, where the polymer-derived substrate can occupy heterogeneous bound configurations and where a single minimum-energy complex can obscure catalytically relevant subpopulations. Together, the calculations show that enhanced activity need not correspond to one universal structural signature: Y87Q primarily decreases the cost of reaching the transition-state region, whereas S187Y preferentially improves stabilizing interactions within that region.

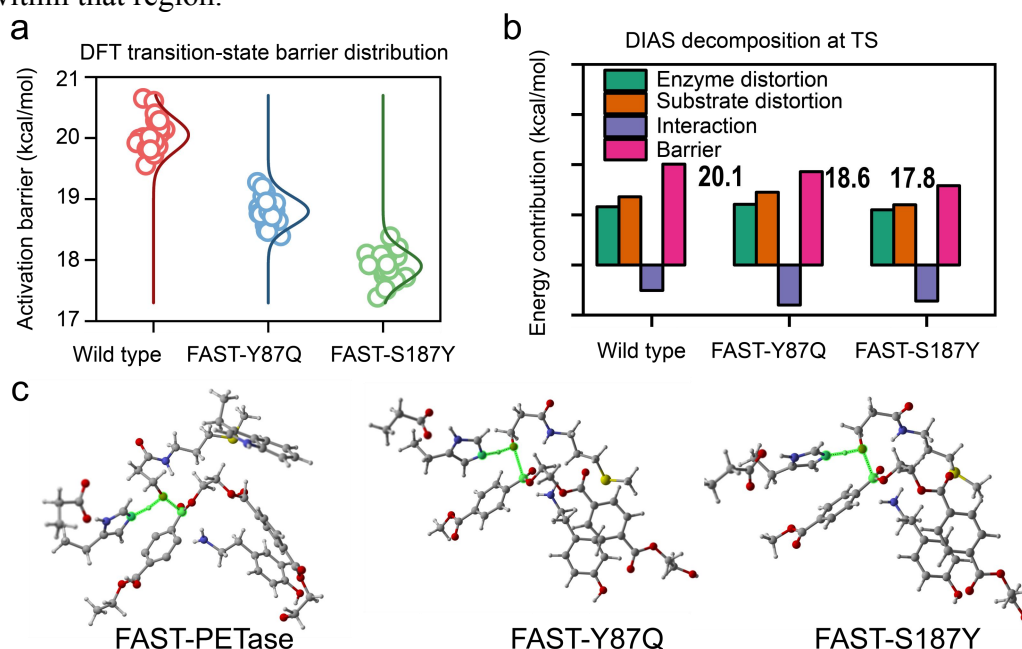


Fig.4 Y87Q and S187Y enhance catalysis through distinct energetic contributions.

(A) DFT-calculated acylation-barrier distributions for FAST-PETase, FAST-Y87Q and FAST-S187Y. (B) Distortion/interaction activation-strain decomposition of the acylation transition state. (C) Representative transition-state structures used for energetic analysis.

2.5 PRS quantity and PRS quality define two separable levels of catalytic preorganization

The contrasting behaviors of Y87Q and S187Y suggest that catalytic preorganization can be separated into two coupled properties (Fig.5). Conventional pre-reaction-state (PRS) or near-attack-conformation analyses classify configurations according to geometric criteria associated with reaction. However, geometrically similar configurations may still differ in oxyanion-hole engagement, proton-transfer organization, substrate distortion, and transition-state stabilization. We therefore distinguish PRS quantity, defined as the probability of accessing a geometrically reaction-ready configuration P_{PRS} from PRS quality, which describes the conditional chemical productivity of configurations within that ensemble. Conceptually, their combined contribution to productive catalytic flux can be expressed. This relation is not intended as a complete kinetic model of PET depolymerization, but rather as a mechanistic separation of two requirements for catalysis.

Within this framework, Y87Q primarily increases PRS quantity by enriching attack-compatible configurations and reducing the structural distortion required to reach the transition-state region. S187Y, in contrast, is consistent with an improvement in PRS quality, with DIAS indicating more favorable interactions during transition-state formation. Thus, comparable increases in PET hydrolysis can arise by improving either access to reaction-ready configurations or their subsequent chemical productivity.

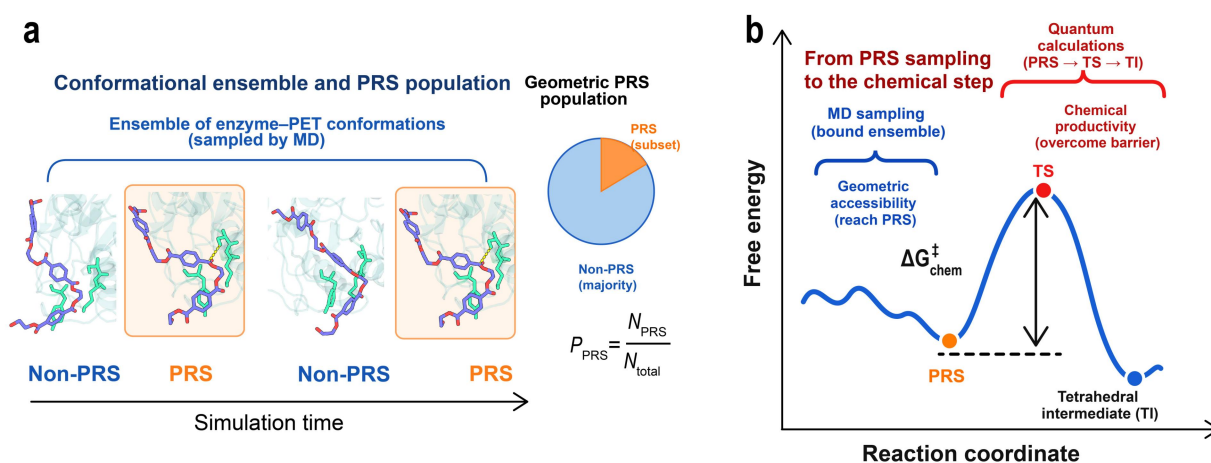


Fig.5 Two-level model of catalytic preorganization based on PRS quantity and quality.

(A) PRS quantity describes the population of geometrically reaction-ready configurations sampled by MD simulations. (B) PRS quality describes the chemical productivity of configurations within the PRS ensemble, as evaluated by progression toward the transition state.

2.6 A bottleneck-guided framework for iterative PET-hydrolase engineering

These observations suggest a bottleneck-guided workflow for PET-hydrolase engineering (Fig.6). Candidate variants may originate from machine learning, structure-based design, docking, or experimental screening, but their discovery scores need not identify the physical origin of improved catalysis. Instead, experimentally validated variants can be evaluated through two mechanistic gates.

Gate 1 evaluates PRS quantity. A low P_{PRS} indicates limited access to reaction-ready geometries and motivates engineering of substrate positioning, local solvation, second-shell interactions, or conformational dynamics. Gate 2 evaluates PRS quality. When suitable geometries are accessible but chemical conversion remains suboptimal, PRS-conditioned structural and quantum-chemical analyses can identify opportunities to improve oxyanion stabilization, proton-transfer organization, electrostatics, or transition-state interactions.

Y87Q and S187Y illustrate these complementary routes: Y87Q predominantly improves geometric accessibility, whereas S187Y is consistent with enhanced chemical stabilization within the reaction-ready ensemble. MD, PRS analysis, and quantum-chemical calculations can therefore be used sequentially to identify the remaining catalytic bottleneck and feed that mechanistic information into the next design cycle.

This framework shifts PRS analysis from a post hoc correlate of activity to a diagnostic step in enzyme engineering. Rather than seeking a single descriptor that predicts all beneficial mutations, it asks which requirement remains limiting: access to a reaction-ready configuration or efficient chemistry once that configuration has been reached.

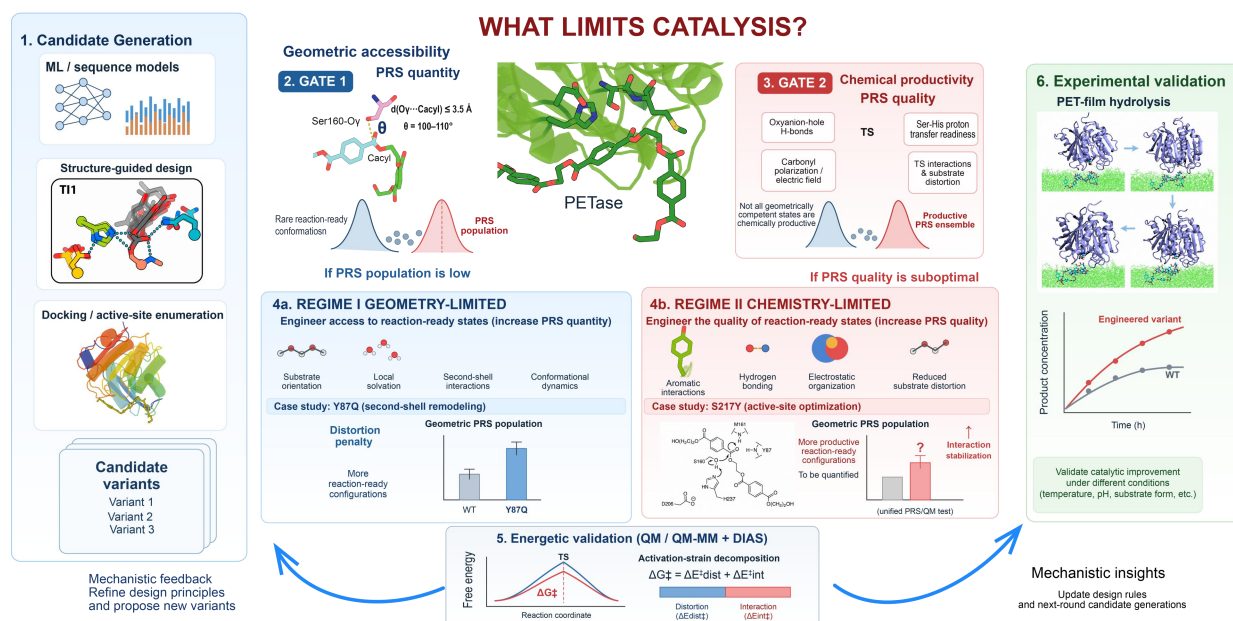


Fig.6 Bottleneck-guided framework for PET-hydrolase engineering.

Candidate variants are evaluated through two mechanistic gates: PRS quantity, defining geometric accessibility, and PRS quality, defining chemical productivity within reaction-ready configurations. Y87Q and S187Y illustrate geometry- and chemistry-limited optimization, respectively; both routes converge on energetic and experimental validation.

3. Discussion

Our results support a two-level view of catalytic preorganization in PET hydrolases. A substrate must first enter an attack-compatible configuration, but geometric competence alone does not ensure efficient chemistry. We therefore distinguish PRS quantity, which describes access to geometric pre-reaction states, from PRS quality, which describes the conditional chemical productivity of configurations within that ensemble. This distinction integrates concepts from near-attack-conformation analysis, conformational preorganization and transition-state stabilization while providing a practical diagnostic for enzyme engineering.

Y87Q exemplifies the first level. The mutation leaves the global fold largely unchanged but reorganizes a local second-shell environment and approximately doubles geometric PRS occupancy. Its reduced distortion contribution in the transition-state analysis provides an energetic counterpart to the MD result: the bound ensemble more frequently reaches structures that require less deformation before chemical reaction. This interpretation is consistent with the broader view that substrate-positioning dynamics can contribute directly to catalytic efficiency. S187Y provides a complementary case. Its activity enhancement is transferable between two scaffolds, yet experimental activity does not follow the docking-energy ranking. The more favorable interaction contribution in DIAS instead points to improved stabilization within the transition-state region. Thus, the two mutations illustrate how comparable improvements in catalytic output can arise from different microscopic bottlenecks.

This framework has a direct design implication. When P_{PRS} is low, optimization should prioritize mutations that reshape substrate positioning, local solvation or second-shell dynamics. When reaction-ready configurations are already accessible, further gains may require improvement of oxyanion-hole engagement, proton-transfer organization, electrostatic complementarity or other interactions that determine PRS quality. Several limitations define the scope of the present interpretation. The simulations begin from enzyme-bound PET-derived substrates and therefore do not explicitly model the full solid-liquid interfacial process. Computed PRS populations and acylation barriers should consequently be interpreted as mechanistic descriptors rather than direct predictors of PET-film degradation rates. In addition, a definitive assignment of S187Y to a PRS-quality-dominated regime would be strengthened by an explicit comparison of its geometric PRS population and PRS-conditioned hydrogen-bond, proton-transfer and electrostatic descriptors. These analyses are discussed further in the Supplementary Information.

More broadly, the framework suggests that geometric sampling and electronic-structure analysis should be treated as successive, complementary levels of the same catalytic problem. It also offers a route to interpreting apparently discordant computational predictions: a favorable binding score, a high PRS population and a low chemical barrier describe different stages of the catalytic process and should not be expected to rank variants identically. Testing PRS quantity and quality across additional PET-hydrolase scaffolds and combinatorial variants will establish how broadly the bottleneck-guided strategy transfers and whether sequential optimization of the two levels can provide a general route from mutation discovery to mechanism-informed enzyme design.

4. Materials and Methods

4.1 Experimental PET-hydrolysis assays

Purified enzymes were evaluated using amorphous PET films. For Y87Q, soluble hydrolysis products were quantified after incubation across 40–70 °C, with time-course measurements performed at 60 °C. For the active-site variant screen, PET films were incubated with purified enzymes at 40 °C for 72 h; soluble products were monitored by absorbance at 260 nm and TPA and MHET were quantified by UPLC. Experimental comparisons used matched conditions within each assay series. Detailed assay information is provided in the Supplementary Information.

4.2 Computational system preparation and molecular dynamics

FAST-PETase, FAST-Y87Q and FAST-S187Y were modeled from the FAST-PETase structure (PDB 7SH6). PET was represented by a trimeric oligomer. Low-energy ligand conformations were generated with CREST/GFN2-xTB [22,23], RESP charges were derived at HF/6-31G(d) [24], and GAFF2/ff19SB parameters were used for PET and protein, respectively [25,26]. Protonation states were evaluated with PROPKA 3.0 [27]. Initial complexes were generated with AutoDock Vina [28,29] and filtered using catalytic geometry before MD simulation. Five independent explicit-solvent replicas were run with Amber 22; equilibrated trajectory regions were used for cross-system statistics and catalytically stable trajectories were extended to 1 μ s [30,31]. Analyses were performed with CPPTRAJ, VMD and PyMOL [32–34].

Geometric PRSs were defined from the Ser160-O γ ...C_{acyl} distance and Bürgi–Dunitz angle. PRS frames were clustered within each MD replica, and four representative configurations per replica were selected, yielding 20 QM cluster models per enzyme. Reactant, transition-state and tetrahedral-intermediate structures were optimized using Gaussian 16 at the ω B97X-D/6-31+G(d) level with SMD water [35–37]; frequency and IRC calculations were used to characterize stationary points and reaction connectivity [38]. IGMH and QTAIM analyses were performed with Multiwfn [39–40], and NBO analysis was used to characterize donor–acceptor interactions. DIAS analysis decomposed activation energies into distortion and interaction contributions using a consistent enzyme/substrate partition [41,42]. Full computational protocols and statistical treatment are provided in the Supplementary Information.

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